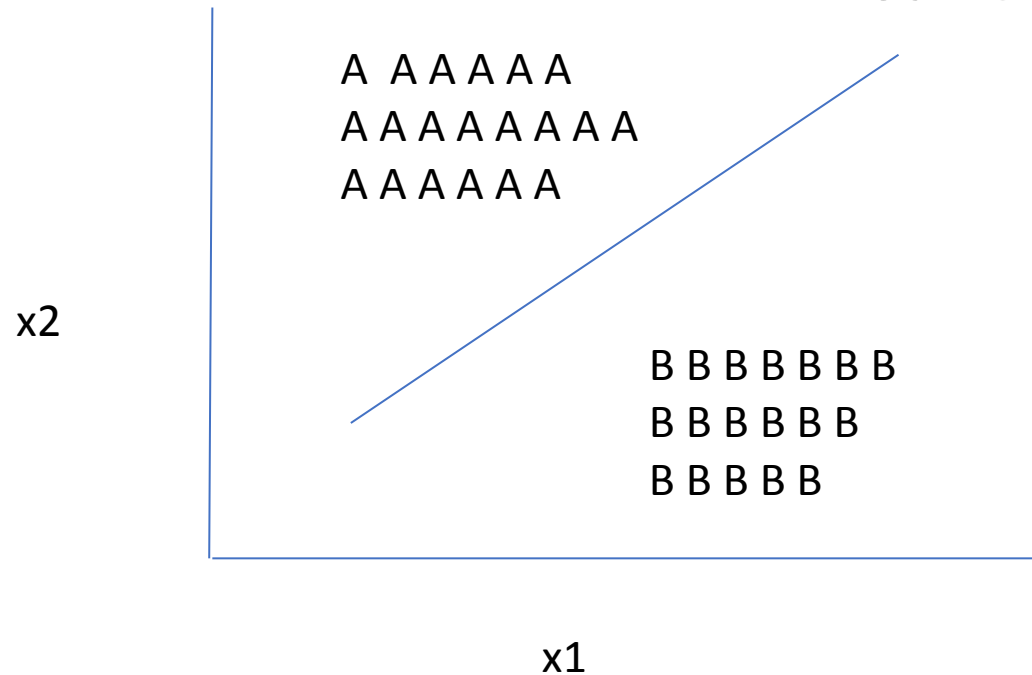


Logistic Regression part 1

Linear Classifier



$$Z = w_1 x_1 + w_2 x_2 + b$$

$$\begin{aligned} Z &= \langle w_1, w_2 \rangle \langle x_1, x_2 \rangle + b \\ &= \mathbf{w} \cdot \mathbf{x} + b \end{aligned}$$

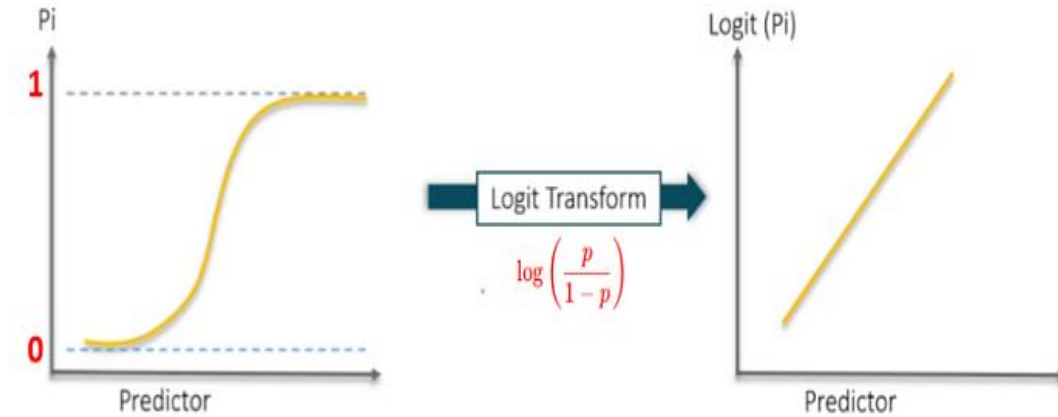
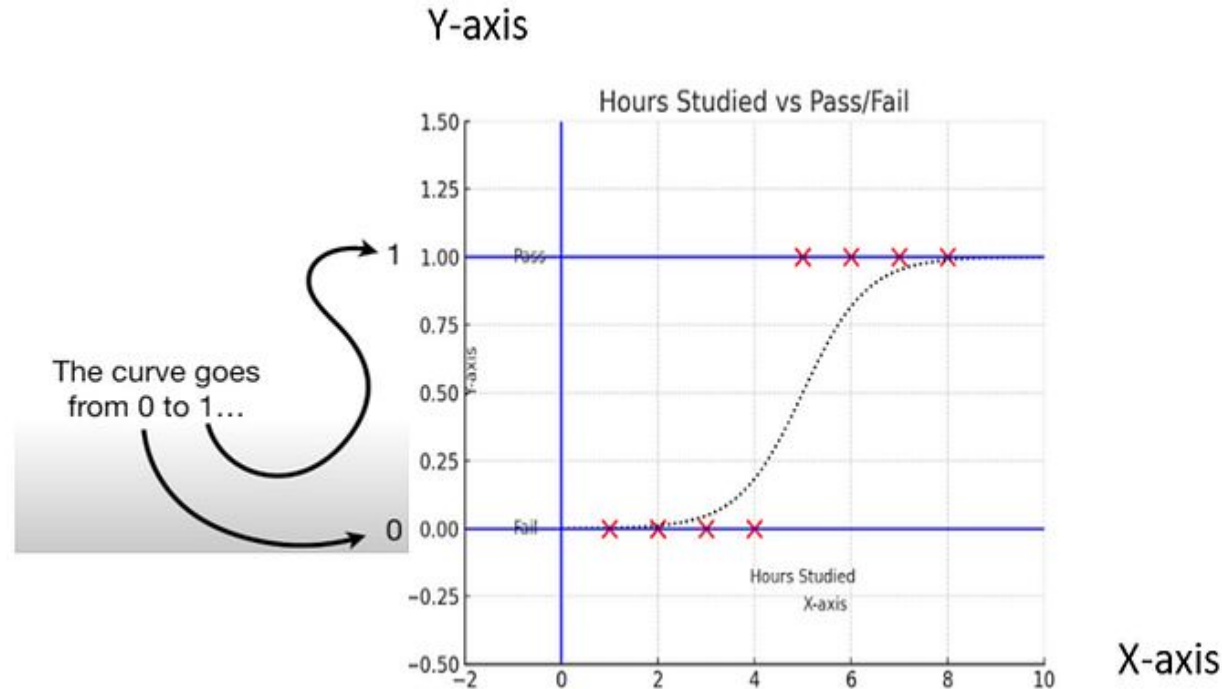
$$\begin{aligned} \text{Where } \mathbf{w} &= \langle w_1, w_2 \rangle \\ \mathbf{x} &= \langle x_1, x_2 \rangle \end{aligned}$$

If we insert x_1 and x_2 values for any object of class A, Z will be positive

If we insert x_1 and x_2 values for any object of class B, Z will be negative

We can say that if $Z > 0$ class is A otherwise class is B

Logistic Regression



When the probability $p = 0$,

$$\log\left(\frac{p}{1-p}\right) = \log\left(\frac{0}{1-0}\right) = \log(0) = -\infty$$

When the probability $p = 1$,

$$\log\left(\frac{p}{1-p}\right) = \log\left(\frac{1}{1-1}\right) = \log(\infty) = +\infty$$

$$P = 0.5 \quad \log(p/p-1) = 0$$

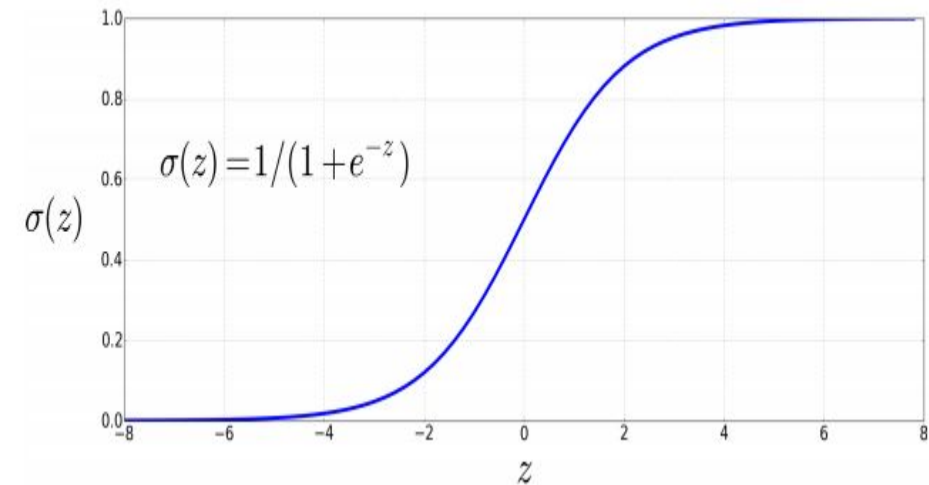
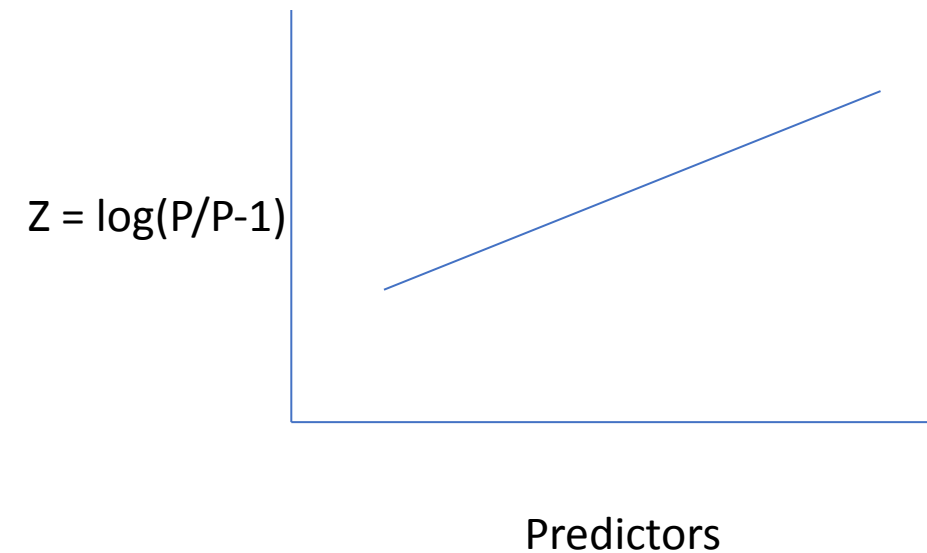
$$P > 0.5 \quad \log(P/P-1) \text{ is +ve}$$

$$P < 0.5 \quad \log(P/P-1) \text{ is -ve}$$

If $Z = \log (P/P-1)$ then $P/P-1 = \exp(Z)$  $P = 1/(1+\exp(-Z))$

We will assume $Z = \log (P/P-1) = W_1X_1 + W_2X_2 + W_3X_3 + \dots + W_n X_n + b = W.X + b$

Where $X_1, X_2, X_3, \dots X_n$ are the feature values and $W_1, W_2, W_3, \dots W_n$ are the weights or coefficients



$$\sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+\exp(-z)}$$

If $z > 0$ then $\sigma(z) > 0.5$

If $z \leq 0$ then $\sigma(z) \leq 0.5$

$$\begin{aligned}
 P(y = 1) &= \sigma(\mathbf{w} \cdot \mathbf{x} + b) \\
 &= \frac{1}{1 + \exp(-(\mathbf{w} \cdot \mathbf{x} + b))}
 \end{aligned}$$

$$\begin{aligned}
 P(y = 0) &= 1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b) \\
 &= 1 - \frac{1}{1 + \exp(-(\mathbf{w} \cdot \mathbf{x} + b))} \\
 &= \frac{\exp(-(\mathbf{w} \cdot \mathbf{x} + b))}{1 + \exp(-(\mathbf{w} \cdot \mathbf{x} + b))}
 \end{aligned}$$

The sigmoid function has the property

$$1 - \sigma(x) = \sigma(-x)$$

Derivative of $\sigma(z) = \sigma(z) (1 - \sigma(z))$

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon words $\in$ doc)	3
x_2	count(negative lexicon words $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	$\ln(\text{word count of doc})$	$\ln(66) = 4.19$

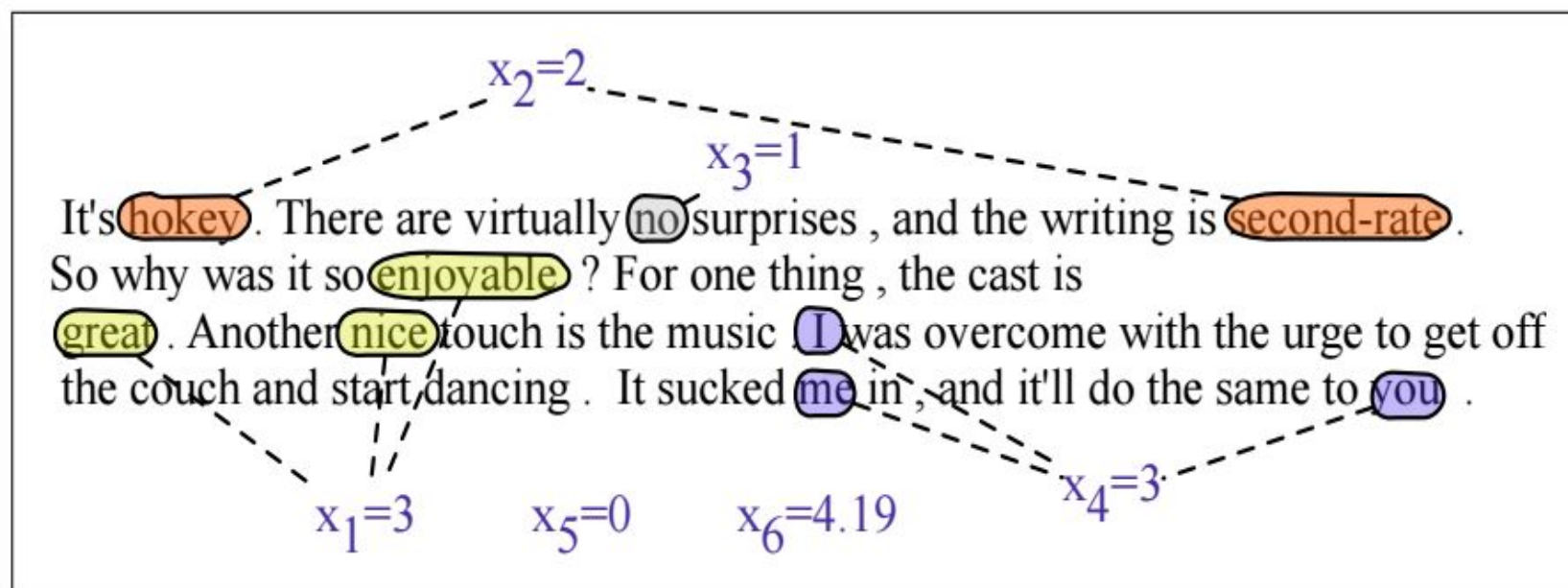


Figure 5.2 A sample mini test document showing the extracted features in the vector x .

$$\begin{aligned}
p(+|x) = P(y = 1|x) &= \sigma(\mathbf{w} \cdot \mathbf{x} + b) \\
&= \sigma([2.5, -5.0, -1.2, 0.5, 2.0, 0.7] \cdot [3, 2, 1, 3, 0, 4.19] + 0.1) \\
&= \sigma(.833) \\
&= 0.70 \\
p(-|x) = P(y = 0|x) &= 1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b) \\
&= 0.30
\end{aligned} \tag{5.8}$$

Scaling input features

$$\begin{aligned}
\mu_i &= \frac{1}{m} \sum_{j=1}^m x_i^{(j)} & \sigma_i &= \sqrt{\frac{1}{m} \sum_{j=1}^m \left(x_i^{(j)} - \mu_i\right)^2} \\
x'_i &= \frac{x_i - \mu_i}{\sigma_i}
\end{aligned}$$

Alternatively, we can **normalize** the input features values to lie between 0 and 1:

$$x'_i = \frac{x_i - \min(x_i)}{\max(x_i) - \min(x_i)} \tag{5.10}$$

Processing multiple examples in matrix form

$$\begin{aligned} \textbf{foreach } x^{(i)} \text{ in input } [x^{(1)}, x^{(2)}, \dots, x^{(m)}] \\ y^{(i)} = \sigma(\mathbf{w} \cdot \mathbf{x}^{(i)} + b) \end{aligned} \quad (5.11)$$

For the first 3 test examples, then, we would be separately computing the predicted $\hat{y}^{(i)}$ as follows:

$$P(y^{(1)} = 1 | x^{(1)}) = \sigma(\mathbf{w} \cdot \mathbf{x}^{(1)} + b)$$

$$P(y^{(2)} = 1 | x^{(2)}) = \sigma(\mathbf{w} \cdot \mathbf{x}^{(2)} + b)$$

$$P(y^{(3)} = 1 | x^{(3)}) = \sigma(\mathbf{w} \cdot \mathbf{x}^{(3)} + b)$$

$$\mathbf{X} = \begin{bmatrix} x_1^{(1)} & x_2^{(1)} & \dots & x_f^{(1)} \\ x_1^{(2)} & x_2^{(2)} & \dots & x_f^{(2)} \\ x_1^{(3)} & x_2^{(3)} & \dots & x_f^{(3)} \\ \dots & & & \end{bmatrix} \quad \mathbf{Y} = \sigma(\mathbf{XW} + \mathbf{b})$$

$$\begin{array}{ccccc} \mathbf{y} & = & \mathbf{X} & \mathbf{w} & + & \mathbf{b} \\ (m \times 1) & & (m \times f) & (f \times 1) & & (m \times 1) \end{array}$$

To derive the parameters $w_1, w_2, w_3, \dots$ and b , we will define a Loss function. We will start with some random values of $w_1, w_2, w_3, \dots$ and b and iteratively update the values using gradient descent until the loss function becomes minimum.

We do this via a loss function that prefers the correct class labels of the training examples to be *more likely*. This is called **conditional maximum likelihood estimation**: we choose the parameters w, b that **maximize the log probability of the true y labels in the training data** given the observations x . The resulting loss function is the **negative log likelihood loss**, generally called the **cross-entropy loss**.

For every example x maximize $P(y = 1 | x)$ or $P(y = 0 | x)$

$$\begin{aligned} \text{Maximize } p(y|x) &= \hat{y}^y (1 - \hat{y})^{1-y} & \text{Maximize } \log p(y|x) &= \log [\hat{y}^y (1 - \hat{y})^{1-y}] \\ & & &= y \log \hat{y} + (1 - y) \log(1 - \hat{y}) \end{aligned}$$

$$L_{\text{CE}}(\hat{y}, y) = -\log p(y|x) = -[y \log \hat{y} + (1-y) \log(1-\hat{y})]$$

$$L_{\text{CE}}(\hat{y}, y) = -[y \log \sigma(\mathbf{w} \cdot \mathbf{x} + b) + (1-y) \log(1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b))]$$

$$\begin{aligned} L_{\text{CE}}(\hat{y}, y) &= -[y \log \sigma(\mathbf{w} \cdot \mathbf{x} + b) + (1-y) \log(1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b))] \\ &= -[\log \sigma(\mathbf{w} \cdot \mathbf{x} + b)] \\ &= -\log(.70) \\ &= .36 \end{aligned}$$

$$\begin{aligned} L_{\text{CE}}(\hat{y}, y) &= -[y \log \sigma(\mathbf{w} \cdot \mathbf{x} + b) + (1-y) \log(1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b))] \\ &= -[\log(1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b))] \\ &= -\log(.30) \\ &= 1.2 \end{aligned}$$

$$L_{\text{CE}}(\hat{y}, y) = -[y \log \sigma(\mathbf{w} \cdot \mathbf{x} + b) + (1 - y) \log (1 - \sigma(\mathbf{w} \cdot \mathbf{x} + b))]$$

To update any parameter w , we will use gradient descent approach.

We need to take gradient of the Loss function with respect to the parameter w

$$w^{t+1} = w^t - \eta \frac{d}{dw} L(f(x; w), y)$$

$$\frac{\partial L_{\text{CE}}(\hat{y}, y)}{\partial w_j} = -(y - \hat{y})x_j$$

$$\begin{aligned} \frac{\partial L_{\text{CE}}(\hat{y}, y)}{\partial w_j} &= [\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y]x_j \\ &= (\hat{y} - y)x_j \end{aligned}$$

For $z = w_1 x_1 + w_2 x_2 + b$

$$w_1 = w_1 - \eta (\hat{y} - y) x_1$$

$$w_2 = w_2 - \eta (\hat{y} - y) x_2$$

$$b = b - \eta (\hat{y} - y)$$



maximize

$$\log P(y|n) = \log [y^y (1-y)^{1-y}]$$

$$= y \log y + (1-y) \log (1-y)$$

$$LCE = -\log P(y|n)$$

$$= -y \log y - (1-y) \log (1-y)$$

$$Z = w_1 x_1 + w_2 x_2 + w_3 x_3 + \dots + b$$

$$w_i = w_i - \eta \frac{\partial LCE}{\partial w_i}$$

$$\frac{\partial LCE}{\partial w_i} = -y \cdot \frac{1}{y} \cdot \frac{\partial}{\partial w_i} \frac{1}{1+e^{-z}}$$

$$- (1-y) \frac{1}{1-y} \frac{\partial}{\partial w_i} (1-y)$$



$$= - \frac{y}{y} \cdot \frac{1}{1+e^{-z}} \left(1 - \frac{1}{1+e^{-z}}\right) \cdot \frac{\partial z}{\partial w_i}$$

$$+ (1-y) \cdot \frac{1}{1-y} \cdot \frac{\partial}{\partial w_i} \frac{1}{1+e^{-z}}$$

$$= - \frac{y}{y} \cdot y (1-y) \cdot n_i$$

$$+ (1-y) \cdot \frac{1}{1-y} \cdot \frac{1}{1+e^{-z}} \left(1 - \frac{1}{1+e^{-z}}\right) \cdot \frac{\partial z}{\partial w_i}$$

$$= -y (1-y) \cdot n_i$$

$$+ (1-y) \frac{1}{1-y} y (1-y) \cdot n_i$$

$$= n_i \{ -y(1-y) + (1-y) \cdot y \}$$

$$= (y - y) n_i$$



$$w_i = u_i - \eta \frac{\partial LCE}{\partial u_i}$$

$$= u_i - \eta \cdot (\hat{y} - y) u_i$$

$$b = b - \eta (\hat{y} - y)$$

$$x_1 = 3 \quad (\text{count of positive lexicon words})$$

$$x_2 = 2 \quad (\text{count of negative lexicon words})$$

Let's assume the initial weights and bias in θ^0 are all set to 0, and the initial learning rate η is 0.1:

$$w_1 = w_2 = b = 0$$

$$\eta = 0.1$$

$$\nabla_{w,b}L = \begin{bmatrix} \frac{\partial L_{\text{CE}}(\hat{y},y)}{\partial w_1} \\ \frac{\partial L_{\text{CE}}(\hat{y},y)}{\partial w_2} \\ \frac{\partial L_{\text{CE}}(\hat{y},y)}{\partial b} \end{bmatrix} = \begin{bmatrix} (\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y)x_1 \\ (\sigma(\mathbf{w} \cdot \mathbf{x} + b) - y)x_2 \\ \sigma(\mathbf{w} \cdot \mathbf{x} + b) - y \end{bmatrix} = \begin{bmatrix} (\sigma(0) - 1)x_1 \\ (\sigma(0) - 1)x_2 \\ \sigma(0) - 1 \end{bmatrix} = \begin{bmatrix} -0.5x_1 \\ -0.5x_2 \\ -0.5 \end{bmatrix} = \begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix}$$

Now that we have a gradient, we compute the new parameter vector θ^1 by moving θ^0 in the opposite direction from the gradient:

$$\theta^1 = \begin{bmatrix} w_1 \\ w_2 \\ b \end{bmatrix} - \eta \begin{bmatrix} -1.5 \\ -1.0 \\ -0.5 \end{bmatrix} = \begin{bmatrix} .15 \\ .1 \\ .05 \end{bmatrix}$$